

SUBAERO: EXHAUSTIVE MASTER TECHNICAL & PHYSICS REFERENCE

End-to-End Documentation: Physics Formulas, ML Models & Detailed Tech Stack

Problem Statement: PS-01 (IIT Indore & Hindustan Aeronautics Limited - HAL)

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Repository: <https://github.com/prajansanjayk1/subaero.git> | **Classification:** Master Technical Reference

1. Complete List of 24 Physics & Thermodynamics Formulas Implemented

Domain / Category	Physics Parameter / Formula Name	Mathematical Equation & Formulation
Atmospheric Gas Dynamics	Speed of Sound (a)	$a = \sqrt{\gamma \cdot R \cdot T_{amb}}$ where $\gamma=1.4$, $R=287.05 \text{ J/(kg}\cdot\text{K)}$
Atmospheric Gas Dynamics	True Airspeed (V)	$V = Mach \cdot a = Mach \cdot \sqrt{\gamma \cdot R \cdot T_{amb}}$
Compressor Thermodynamics	Compressor Pressure Ratio	$PR_{comp} = P_3 / P_2$
Compressor Thermodynamics	Ideal Isentropic Exit Temp	$T_{2_is} = T_{amb} \cdot (P_3 / P_2)^{((\gamma-1)/\gamma)}$
Compressor Thermodynamics	Isentropic Efficiency (η_c)	$\eta_c = (T_{2_is} - T_{amb}) / (T_2 - T_{amb})$
Compressor Thermodynamics	Compressor Temp Rise	$dT_{comp} = T_3 - T_2 \text{ [Kelvin]}$
Compressor Thermodynamics	Compressor Specific Work	$w_{comp} = c_p \cdot (T_3 - T_2)$ where $c_p=1005 \text{ J/(kg}\cdot\text{K)}$
Combustor Thermal State	Combustor Temp Ratio	$TR_{comb} = T_3 / T_2$
Combustor Thermal State	Heat Addition Rate (q_{in})	$q_{in} = c_p \cdot (T_3 - T_2) \text{ [J/kg]}$
Combustor Thermal State	Fuel-Air Ratio Proxy (f)	$f = FuelFlow / (RPM \cdot P_2 / T_2)$
Turbine Work Dynamics	Turbine Temp Expansion Ratio	$TR_{turb} = T_4 / T_3$
Turbine Work Dynamics	Turbine Pressure Expansion Ratio	$PR_{turb} = P_3 / P_4$
Turbine Work Dynamics	Work Coefficient (W)	$W = (T_3 - T_4) / T_3$
Turbine Work Dynamics	Turbine Temp Drop	$dT_{turb} = T_3 - T_4 \text{ [Kelvin]}$
Thermal Stress & Limits	Thermal Stress Index (σ)	$\sigma_{thermal} = (T_3 / T_2) \cdot (P_3 / P_2)$
Thermal Stress & Limits	EGT Safety Margin	$EGT_{margin} = 1273.15 - T_4 \text{ [Kelvin]}$
Performance Surrogates	Net Thrust Force (F_{net})	$F_{net} = \dot{\phi}(z, W_{thrust}) \text{ [Newtons]}$
Performance Surrogates	Fuel TSFC	$TSFC = FuelFlow / F_{net} \text{ [g/(N}\cdot\text{s)]}$
Physics Guardian Rules	Compressor Work Rule	$T_3 > T_2$ (Flag Compressor_Work_Inversion if violated)
Physics Guardian Rules	Combustor Expansion Rule	$T_4 < T_3$ (Flag Combustor_Heat_Inversion if violated)
Physics Guardian Rules	Pressure Ratio Rule	$P_3 > P_2 \cdot 1.05$ (Flag Compression_Loss_Surge if violated)
Physics Guardian Rules	EGT Overheat Limit	$T_4 \leq 1273.15 \text{ K}$ (Flag EGT_CRITICAL_OVERTEMP if violated)
Health Tree & RUL	11-Node Health Tree Aggregation	$H_{overall} = 0.40 \cdot H_{mech} + 0.25 \cdot H_{thermal} + 0.20 \cdot H_{press} + 0.10 \cdot H_{comb} + 0.05 \cdot H_{eff}$
Health Tree & RUL	RUL Degradation Trajectory	$RUL = (EMA10(t) - 0.70) / (dH/dt + 1e-9) \text{ [Cycles]}$

2. Machine Learning Models Specification (Detailed Version)

Primary ML Architecture: 100% White-Box Regularized Polynomial Matrix Model Engine

- **Surrogate Class:** Scikit-Learn Degree-2 Regularized Polynomial Ridge Regression (alpha = 50.0).
- **Mathematical Matrix Formulation:** $y_{\text{hat}} = w_0 + \sum_{k=1}^{91} w_k * \phi_k(z_1, \dots, z_{12})$
- **Polynomial Term Expansion (91 terms):** 1 Bias Intercept, 12 Linear Terms, 12 Quadratic Terms (z_i^2), and 66 Cross-Sensor Interaction Terms ($z_i * z_j$).
- **Client Asset Serialization:** Compiled into lightweight JSON asset src/assets/whitebox_models_12sensors.json (15.2 KB) for Oms browser inference.
- **Python Model Files:** Serialized Scikit-Learn pipelines stored in trained_models_whitebox_12s/.
- **RUL Predictor Engine:** Sensor-based 10-cycle Exponential Moving Average (EMA_10) degradation rate velocity extrapolator.
- **Black-Box Purge:** 0% Tree ensembles (LightGBM, XGBoost, CatBoost, ExtraTrees completely removed).

Target Parameter	ML Model Algorithm	Accuracy	MAE / RMSE Metric	R² Score
TSFC (Fuel Consumption)	Degree-2 Polynomial Ridge	99.97%	0.000253 g/(N*s) 0.000339	0.9969
Thrust Force (N)	Degree-2 Polynomial Ridge	99.27%	439.98 N 572.33 N	0.9989
Combustor Health	Degree-2 Polynomial Ridge	98.98%	0.010233 0.013416	0.7276
Overall Engine Health	Degree-2 Polynomial Ridge	98.77%	0.012305 0.016335	0.8808
Compressor Health	Degree-2 Polynomial Ridge	98.30%	0.016984 0.023027	0.8816
Turbine Health	Degree-2 Polynomial Ridge	98.02%	0.019784 0.025685	0.7392

3. Complete Technology Stack Specification (Detailed Versions)

Tech Stack Component Layer	Framework / Technology Package	Exact Version	Engineering Purpose & Role
Machine Learning & Data Science	Python Language Runtime	v3.11.7	Core ML pipeline training, data processing, and CLI predictors.
Machine Learning & Data Science	Scikit-Learn	v1.4.0	PolynomialFeatures, Ridge regression, and StandardScaler pipelines.
Machine Learning & Data Science	NumPy	v1.26.4	High-performance matrix operations and polynomial term dot products.
Machine Learning & Data Science	Pandas	v2.2.0	Dataset manipulation, CSV telemetry parsing, and ground truth merging.
Machine Learning & Data Science	Joblib	v1.3.2	Serialization and deserialization of trained white-box ML models.
Machine Learning & Data Science	SciPy & CoolProp	v1.12.0 / v6.4.3	Thermodynamic fluid properties and Brayton cycle gas dynamics calculations.
Frontend UI & Web App	TypeScript	v5.2.2	Type-safe frontend logic and client-side Oms matrix prediction engine.
Frontend UI & Web App	React Framework	v18.2.0	Component-driven Mission Control Dashboard UI.
Frontend UI & Web App	Vite Build Engine	v6.4.3	Lightning-fast HMR dev server and production bundling.
Frontend UI & Web App	Tailwind CSS	v3.4.1	Modern glassmorphism UI styling and responsive dashboard layout.
Frontend UI & Web App	Recharts & Lucide-React	v2.12.2 / v0.344.0	Telemetry sparklines, health accuracy charts, and aerospace UI icons.
3D Digital Twin Engine	Unity Engine & Blender	2022.3 LTS / 4.0 LTS	3D Turbojet CAD modeling and real-time casing thermal heatmap viewport.
Backend & Database	FastAPI & Uvicorn	v0.109.0 / v0.27.0	Asynchronous REST API gateway for edge telemetry streaming.
Documentation & Publishing	ReportLab & pdfLaTeX	v4.1.0 / MiKTeX 24.1	PDF document compilation and official 15-page LaTeX technical report.